

Module-03 (STLC)

Q1. Create a Test Plan for an Online Shopping Website.

Ans:

Project Name	Online Shopping Website
Document Purpose	Defines the scope, approach, resources, and schedule of testing activities for the Online Shopping Website.
Objective	● Verify that all core features (browsing, registration, login, cart, checkout, payment) work as per requirements. ● Identify defects before release and ensure a smooth, secure shopping experience.
Scope - In Scope	● User Registration & Login ● Product Search, Listing & Filtering ● Shopping Cart (Add/Update/Remove items) ● Checkout & Payment ● Order Confirmation & Order History
Scope - Out of Scope	● Admin/backend panel testing ● Third-party payment gateway internal testing ● Load testing beyond basic performance checks
Test Types	● Functional Testing ● UI/Usability Testing ● Compatibility Testing (browsers/devices) ● Security Testing (basic - input validation, SQL injection, XSS) ● Regression Testing ● Performance Testing (basic page-load checks)
Test Environment	● Browsers: Chrome, Firefox, Edge ● Devices: Desktop and Mobile (responsive view) ● Test Environment: Staging/QA build with test database ● Test Data: Valid/invalid user accounts, sample products, test payment cards
Entry Criteria	● Requirements/user stories are finalized and shared ● Test build is deployed on the QA environment ● Test cases are reviewed and approved
Exit Criteria	● All planned test cases executed ● No open Critical/High severity defects ● Test summary report shared with stakeholders
Roles & Responsibilities	● QA Engineer: Test case design, execution, defect reporting ● Developer: Defect fixing ● Project Manager: Sign-off and release approval

Deliverables	• Test Plan • Test Cases / Checklist • Defect Reports • Test Summary Report
Tools	Manual testing with Excel/Google Sheets for test cases; browser DevTools for basic checks; bug tracking tool (e.g., Jira) for defect logging.
Risks & Mitigation	• Risk: Unclear requirements → Mitigation: Clarify with BA before test design • Risk: Limited test time → Mitigation: Prioritize Critical/High test cases first • Risk: Unstable test environment → Mitigation: Verify environment before execution begins

Q2. Prepare a Test Checklist.

Ans:

2.1 Login Checklist

Test Area	Validation Point	Expected Result
UI	All login page fields and buttons (Username, Password, Login, Forgot Password) are displayed correctly	All elements are visible and correctly labeled
Field Validation	Username field is mandatory	Error message shown when left empty
Field Validation	Password field is mandatory	Error message shown when left empty
Field Validation	Both username and password left empty and submitted	Appropriate error message displayed; form not submitted
Functional	Login with valid username and valid password	User logs in successfully and is redirected to the home/dashboard page
Functional	Login with invalid username and valid password	Generic error message displayed; login rejected
Functional	Login with valid username and invalid password	Generic error message displayed; login rejected
Functional	Login with invalid username and invalid password	Generic error message displayed; login rejected

Test Area	Validation Point	Expected Result
Security	Username field is case-sensitive as per business rule	Behavior matches defined rule (accept/reject mismatched case)
Security	Password field masks entered characters	Characters shown as dots/asterisks, not plain text
UI	Show/Hide password toggle (eye icon) works correctly	Password text visibility toggles as expected
Security	SQL injection strings entered in username/password	Input treated as literal text; login rejected with standard error, no application error
Security	Script/HTML tags entered in username/password (XSS check)	Input is sanitized/escaped; no script executes
Boundary	Username/password with maximum allowed character length	Field accepts input up to the defined limit; excess characters blocked
Negative	Multiple failed login attempts with wrong password	Account is locked/warned as per defined threshold, or standard error shown each time
Functional	'Forgot Password' link navigates to the password recovery page	User is redirected to the correct recovery page
Functional	'Remember Me' checkbox retains session on next visit	User remains logged in on return visit when checked
Functional	Session expires after defined inactivity period	User is logged out and redirected to login page
Functional	Logout option ends the session correctly	User is logged out and redirected to login/home page; back button does not restore session
Performance	Login page and post-login redirect load within acceptable time	Page loads without noticeable delay

2.2 Registration Checklist

Test Area	Validation Point	Expected Result
UI	All registration fields (Name, Email, Mobile, Password, Confirm Password, T&C) are displayed	All fields and labels are visible and correctly placed
Field Validation	All mandatory fields are validated on submit	Error messages shown for each empty mandatory field
Field Validation	Valid email format is accepted	Registration proceeds without error

Test Area	Validation Point	Expected Result
Field Validation	Invalid email format (e.g., missing '@' or domain) is rejected	Appropriate error message displayed
Functional	Registration attempted with an already registered email	Error message shown indicating the account/email already exists
Field Validation	Password meets minimum strength rules (length, character mix)	Weak passwords are rejected with a clear validation message
Field Validation	Confirm Password must match Password field	Error shown when values do not match
Field Validation	Mobile number accepts only valid numeric format and length	Non-numeric or incorrect-length input is rejected
Functional	Terms & Conditions checkbox must be selected before submission	Form cannot be submitted unless checkbox is checked
Field Validation	Name field rejects numbers and special characters (if restricted)	Invalid characters trigger a validation error
Negative	Form submitted with all fields empty	Validation errors shown for each mandatory field; no request sent to server
Functional	Successful registration redirects user appropriately	User is redirected to login page or dashboard with a success message
Functional	Confirmation email or OTP is sent after successful registration	Email/OTP is received and matches the entered account details
Boundary	Name/Email/Password fields tested with minimum and maximum allowed length	Fields accept values within limits and reject values outside limits
Security	SQL injection or script input entered in text fields	Input is sanitized; no application error or script execution occurs
Functional	Duplicate form submission is prevented (e.g., double-click on Register button)	Only one registration request is processed
UI	Inline validation messages appear near the relevant field	Error messages are clear and correctly positioned
Functional	Cancel/Back navigation from registration page	User is redirected without partial/invalid data being saved
Functional	Password field masks input by default with a show/hide toggle	Characters are hidden by default; toggle reveals/hides correctly
Compatibility	Registration form works correctly across supported browsers/devices	Form renders and functions consistently on all tested platforms

2.3 Checkout Checklist

Test Area	Validation Point	Expected Result
UI	Cart page displays correct product name, image, price, and quantity	All item details match what was added to the cart
Functional	Increasing/decreasing item quantity updates price and cart total	Line item price and cart total recalculate correctly
Functional	Removing an item from the cart updates the cart contents and total	Item is removed and total is recalculated correctly
Negative	Proceeding to checkout with an empty cart	System prevents checkout or shows an appropriate empty-cart message
Field Validation	Shipping address form validates mandatory fields (name, address, city, zip, phone)	Error messages shown for missing/invalid fields
Functional	User can select from previously saved addresses	Selected address is correctly applied to the order
Functional	'Billing address same as shipping' option works correctly	Billing fields auto-fill and match shipping address when selected
Functional	All supported payment methods (Card, COD, Wallet, etc.) are listed and selectable	Each payment option can be selected and proceeds correctly
Field Validation	Card number field accepts only valid numeric card formats	Invalid card numbers are rejected with an error message
Field Validation	Invalid/expired card details are rejected	Error message displayed; order is not placed
Field Validation	CVV and expiry date fields are validated for correct format	Invalid CVV/expiry triggers a validation error
Functional	Valid coupon/promo code is applied at checkout	Discount is correctly reflected in the order summary and total
Negative	Invalid or expired coupon code is entered	Error message shown; discount is not applied
Functional	Order summary (items, quantity, price, tax, shipping, total) matches the cart	All values in the summary match the cart before payment
Functional	Tax and shipping charges are calculated correctly based on address/order value	Charges match the defined business rules
Functional	Clicking 'Place Order' with valid details completes the order	Order confirmation page/message is displayed with an order ID
Functional	Order confirmation email/SMS is sent after successful checkout	Confirmation is received with correct order details

Test Area	Validation Point	Expected Result
Negative	Payment failure (e.g., declined card) during checkout	Clear error message shown; user can retry payment without losing cart data
Functional	Guest checkout (without account login), if supported	User can complete checkout without registering an account
Functional	Navigating back during checkout does not lose cart contents	Cart items remain intact when user returns to a previous step